

Dividing Complex Numbers with Conjugates

Answer Key

Algebra 2

Quick Answers

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|-------------|--------------|---------------------|
| 1. 53 | 5. $4 + 5i$ | 9. $2 + 3i$ |
| 2. $2 - 3i$ | 6. $1 - 4i$ | 10. $2 + i, 2 + 4i$ |
| 3. $3 + i$ | 7. $-2 - 3i$ | |
| 4. $2 + 3i$ | 8. $3 - 4i$ | |
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Curriculum

Level: Algebra 2

HSN-CN.A.3 – problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10

Difficulty 1–5 of 5 across 11 tagged checks.

Common wrong answers

1. If they got 45: treated i squared as $+1$, so the last product read 49 minus 4 instead of 49 plus 4
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1. Multiply $7 + 2i$ by its conjugate.

The conjugate keeps the real part and flips the sign of the imaginary part, so it is $7 - 2i$. Multiply the two:

$$(7 + 2i)(7 - 2i) = 49 - 14i + 14i - 4i^2 = 49 - 4(-1) = 49 + 4$$

The middle terms cancel and the $-4i^2$ turns into $+4$, which is why the product is real — this is the fact the whole worksheet runs on.

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2. Simplify $\frac{6 + 4i}{2i}$.

The conjugate of $2i$ is $-2i$. Multiply the top and the bottom by it:

$$\frac{(6 + 4i)(-2i)}{(2i)(-2i)} = \frac{-12i - 8i^2}{-4i^2} = \frac{-12i + 8}{4} = \frac{8 - 12i}{4}$$

Divide both parts of the numerator by 4.

$2 - 3i$

Grading note: $-2 + 3i$ is the diagnostic wrong answer. It comes from using $1/i = i$; the truth is $1/i = -i$.

3. Simplify $\frac{5+5i}{2+i}$.

Multiply the top and the bottom by the conjugate $2-i$:

$$\frac{(5+5i)(2-i)}{(2+i)(2-i)} = \frac{10-5i+10i-5i^2}{4+1} = \frac{10+5i+5}{5} = \frac{15+5i}{5}$$

$$\boxed{3+i}$$

Check: $(3+i)(2+i) = 6+3i+2i-1 = 5+5i$ ✓

4. Simplify $\frac{11+10i}{4-i}$.

The denominator is $4-i$, so its conjugate is $4+i$:

$$\frac{(11+10i)(4+i)}{(4-i)(4+i)} = \frac{44+11i+40i+10i^2}{16+1} = \frac{44+51i-10}{17} = \frac{34+51i}{17}$$

Divide both parts by 17: $34/17 = 2$ and $51/17 = 3$.

$$\boxed{2+3i}$$

5. Impedance from voltage and current.

Given $V = 2+23i$ volts and $I = 3+2i$ amps, impedance is $Z = V/I$, so the computation is a division by the current.

$$Z = \frac{2+23i}{3+2i} = \frac{(2+23i)(3-2i)}{(3+2i)(3-2i)} = \frac{6-4i+69i-46i^2}{9+4} = \frac{6+65i+46}{13} = \frac{52+65i}{13}$$

Divide both parts by 13.

$$\boxed{4+5i \text{ ohms}}$$

Check: $(4+5i)(3+2i) = 12+8i+15i-10 = 2+23i$ volts ✓. The real part is the resistance and the imaginary part is the reactance, both in ohms.

6. Simplify $\frac{11-10i}{3+2i}$, **and answer Priya.**

Dividing part by part is not a legal move: division is defined by multiplication, and $(\frac{11}{3}-5i)(3+2i)$ is nowhere near $11-10i$. Only multiplying the top and the bottom by the conjugate keeps the value of the quotient unchanged.

$$\frac{(11-10i)(3-2i)}{(3+2i)(3-2i)} = \frac{33-22i-30i+20i^2}{9+4} = \frac{33-52i-20}{13} = \frac{13-52i}{13}$$

$$\boxed{1-4i}$$

Grading note: $3.67-5i$ is the part-by-part answer, the error this problem was built to expose.

7. Simplify $\frac{9-6i}{3i}$.

The conjugate of $3i$ is $-3i$:

$$\frac{(9-6i)(-3i)}{(3i)(-3i)} = \frac{-27i+18i^2}{-9i^2} = \frac{-18-27i}{9}$$

Divide both parts by 9. Both parts come out negative, so keep both signs.

$$\boxed{-2-3i}$$

8. Simplify $\frac{25}{3+4i}$.

First the requested statement: $(3+4i)(3-4i) = 9+16 = 25$. The denominator times its conjugate is 25, the same as the numerator.

$$\frac{25(3-4i)}{(3+4i)(3-4i)} = \frac{25(3-4i)}{25} = 3-4i$$

$$\boxed{3-4i}$$

When the numerator is real, dividing by $a+bi$ just multiplies by the conjugate and scales by a^2+b^2 — here the scaling was exactly 1.

9. Current through the speaker.

From $V = I \cdot Z$, dividing both sides by Z gives $I = V/Z$, with $V = 1+8i$ volts and $Z = 2+i$ ohms.

$$I = \frac{1+8i}{2+i} = \frac{(1+8i)(2-i)}{(2+i)(2-i)} = \frac{2-i+16i-8i^2}{4+1} = \frac{2+15i+8}{5} = \frac{10+15i}{5}$$

$$\boxed{2+3i \text{ amps}}$$

Check: $(2+3i)(2+i) = 4+2i+6i-3 = 1+8i$ volts ✓

10. Synthesis: two divisions in disguise.

(a) Isolating x means dividing by $3+2i$, so multiply by the conjugate $3-2i$:

$$x = \frac{4+7i}{3+2i} = \frac{(4+7i)(3-2i)}{9+4} = \frac{12-8i+21i+14}{13} = \frac{26+13i}{13}$$

$$\boxed{x = 2+i}$$

(b) A zero is where $f(x) = 0$, so $(1-i)x = 6+2i$ and again the unknown is freed by a division — this time by $1-i$, whose conjugate is $1+i$:

$$x = \frac{6+2i}{1-i} = \frac{(6+2i)(1+i)}{1+1} = \frac{6+6i+2i-2}{2} = \frac{4+8i}{2}$$

$$\boxed{x = 2+4i}$$

The conjugate used in (b) is $1+i$, because the denominator was $1-i$ and the conjugate is what makes the denominator real: $(1-i)(1+i) = 1+1 = 2$.